From Queue to Graph:

Diagnosing Structural Bottlenecks in Indian High Courts

Joy Bose

Senior Data Scientist and Independent Researcher, Bengaluru

joy.bose@ieee.org

Abstract

Studies of judicial pendency in India treat the court system as a single queue: cases arrive, cases are disposed, and the backlog grows or shrinks. This approach produces alarming numbers but no actionable diagnosis, because it treats all courts as instances of the same failing process. They are not.

We analyse 2.45 million writ petition records across ten Indian High Courts, drawn from the DAKSH High Court Data Portal and supplementary judgment datasets. We show that Indian High Courts fail in structurally different ways at different stages of the case lifecycle. Six courts show an Input bottleneck: cases pile up at the admission stage and are never scheduled for hearing. Two show a Capacity bottleneck: cases are admitted but there is no bench time to hear them. Two show an Output bottleneck: cases have been fully argued but written judgments have not been delivered.

Median case duration varies 26-fold across courts, from 146 days in Chhattisgarh to 3,792 days in Calcutta. Both operate under the same Constitution and the same procedural code. Stage vocabulary, a proxy for administrative standardisation, ranges from 16 labels in Meghalaya to 350 in Calcutta. We formalise this variation as a Semantic Disorder Index, which captures registry encoding practice rather than court performance.

We also show that the primary publicly available dataset for Indian High Court research contains systematic data quality failures that make cross-court comparison difficult without careful handling. These failures are not incidental. They reflect the same institutional heterogeneity documented in the case flow data.

Our main policy conclusion is that a national policy of increasing judicial strength will help Input and Capacity bottleneck courts but will have no effect on Output bottleneck courts, where the constraint is judgment delivery rather than bench availability. Effective reform requires court-specific diagnosis before court-specific prescription.

1. Introduction

India's courts carry approximately 44 million pending cases. The number is cited so often it has become background noise. What is less often asked is whether all those cases are pending for the same reason.

The standard answer, implicit in most policy discussion, is yes. Cases pile up because there are not enough judges, not enough courtrooms, and too many adjournments. The solution is therefore to appoint more judges, build more courtrooms, and crack down on adjournments. This prescription is applied nationally, to all courts simultaneously.

This paper argues that the standard answer is wrong, or at least incomplete. Courts do not all fail in the same way. Some courts fail at the front end: cases are filed and registered but never scheduled for a hearing. Others fail in the middle: cases are heard repeatedly but nothing happens between hearings. Still others fail at the back end: arguments are complete and the judge has reserved the case for a written judgment, but the judgment is not delivered for months or years.

These are three different problems. They require three different solutions. A policy that works for one type of court may do nothing for another, and may even make things worse by diverting attention and resources from the actual constraint.

To demonstrate this, we analyse writ petition records from ten Indian High Courts. Writ petitions are the most policy-relevant category of High Court work. They include challenges to government orders, service disputes, land acquisition matters, and fundamental rights cases. They are also the category best covered by the DAKSH High Court Data Portal, which is the main publicly available source of case-level data for Indian High Courts. The analysis covers writ petitions only; civil suits, criminal trials, and tribunal matters may exhibit different bottleneck patterns and are left for future work.

We find that courts cluster into three structural types based on where in the case lifecycle cases are accumulating. We measure how long cases take in courts where reliable date data is available, and find a 26-fold difference between the fastest and slowest court. We examine the vocabulary that courts use to describe hearing stages, and find that some courts use 16 labels while others use 350, a difference that reflects institutional complexity and recording practice rather than performance.

We also find significant problems in the underlying data that affect any analysis. Two courts have a data artifact that makes duration analysis impossible. One court's case file uses a non-standard text encoding because it includes regional language text. Several courts use stage labels that are specific to their own registry and cannot be compared with other courts without manual interpretation. We document these problems explicitly because honest description of data quality is itself a contribution to the field.

The paper is structured as follows. Section 2 reviews the existing literature. Section 3 describes the data. Section 4 describes the methods. Section 5 presents findings for all ten courts including a sensitivity analysis. Section 6 discusses the COVID shock evidence and case-type findings from a supplementary dataset. Section 7 draws policy conclusions.

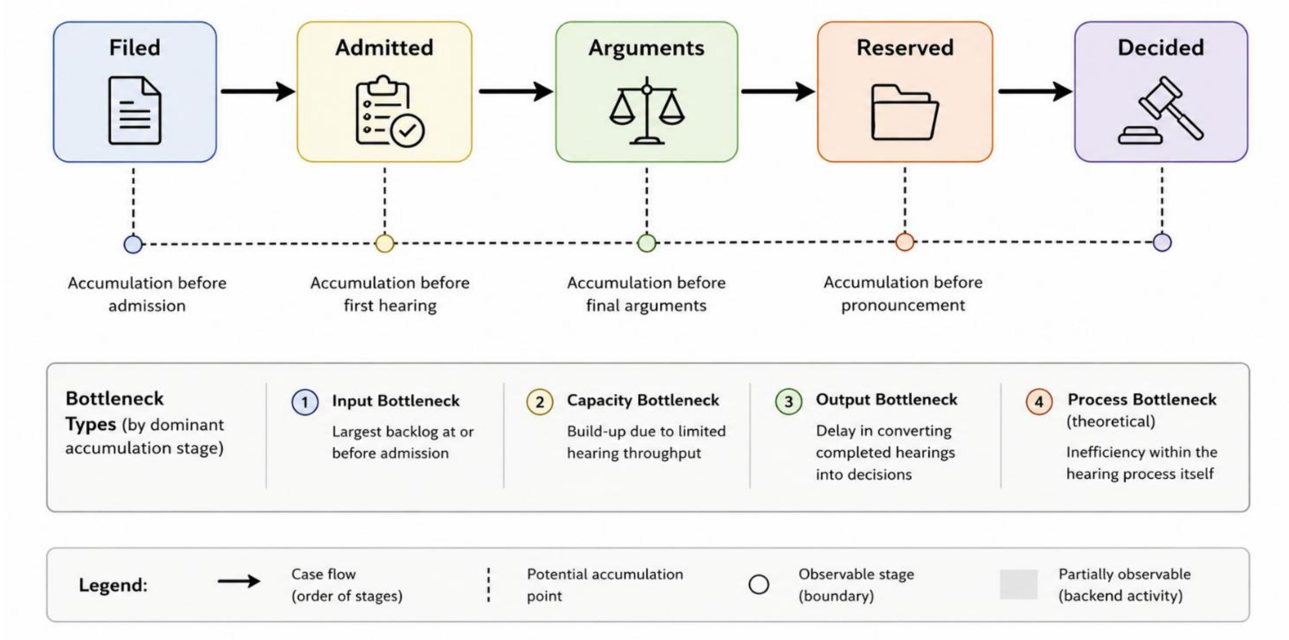

Figure 1: Stages and bottlenecks in a typical court case lifecycle

2. Prior Work

2.1 Quantitative Studies of Judicial Delay

Research on Indian judicial delay has grown significantly since the National Judicial Data Grid (NJDG) made aggregate pendency statistics publicly available. The most commonly cited recent work is Verma et al. (2023), which uses linear regression on HC-NJDG data to project clearance times for each High Court. That study finds that some courts will take over a century to clear their backlogs at current disposal rates, and that pendency is growing faster than disposal in most courts.

Verma (2023a) conducted the first longitudinal analysis of HC-NJDG data, collecting statistics across 73 days between August 2017 and December 2018. That study documented serious errors in judge count data, irregular update practices by several courts, and a large gap between cases listed and cases disposed on any given day. Our work differs in two fundamental ways: we use case-level DAKSH data rather than aggregate NJDG dashboards, and we classify courts by procedural stage bottleneck rather than by aggregate disposal rates. Notably, the NJDG data quality failures Verma documented persist in the DAKSH data we analyse from 2021, suggesting these problems are structural rather than transitional.

This type of analysis is useful for documenting the scale of the problem. It is less useful for diagnosing its causes. Linear regression on aggregate counts treats the court as a black box: inputs and outputs are measured, but the process inside is not. It cannot distinguish between a court that is failing because too many cases arrive at once, a court that is failing because cases move too slowly through the hearing process, and a court that is failing because judges are sitting on reserved judgments. Each of these requires a different response.

Earlier quantitative work includes studies by Daksh India (2016) and the NIPFP (National Institute of Public Finance and Policy) on judicial productivity and its determinants. These studies identify judicial vacancies, case type composition, and state-level economic development as correlates of disposal rates. Our work builds on this tradition but shifts from correlation to structural classification.

A closely related contribution is Bakshi, Kim, and Randhawa (2025), who apply a queueing-theoretic simulation framework to Supreme Court of India data. They find that strategic reallocation of judicial time from preadmission screening to postadmission merit hearings can reduce average resolution time by 65% without any increase in judicial headcount. This provides empirical support for our argument that vacancy filling is not the only or even the primary lever for reducing delay.

2.2 Legal and Policy Literature

The legal literature on judicial delay in India is extensive. The Law Commission of India has addressed the problem in multiple reports, most extensively in its 245th Report (2014) on arrears and backlog. The report identifies adjournments, vacancies, and infrastructure deficits as primary causes, and recommends case management reform, increased judicial strength, and mandatory pre-litigation mediation.

Supreme Court judgments including Hussainara Khatoon (1979) and P. Ramachandra Rao (2002) have established that speedy justice is a fundamental right under Article 21 of the Constitution. More recently, the Supreme Court has directed the NJDG project to improve data collection and transparency.

Moog (2016) identifies that the persistence of judicial delay in India reflects misaligned incentives across three groups: judges who prioritise administrative job security, lawyers who benefit financially from prolonged proceedings, and registry staff who control cause-list placement informally. These structural incentive failures help explain why legislative reforms designed to cap adjournments repeatedly fail in practice.

What is missing from both the quantitative and legal literature is a systematic characterisation of how different courts fail differently. The Law Commission treats delay as a single phenomenon. Quantitative studies aggregate across courts or treat court fixed effects as unexplained noise. Neither approach asks the structural question: at which stage of the case lifecycle is each court failing, and why?

2.3 Computational Legal Studies and Data Quality

The use of computational methods in Indian legal research is growing. Work in natural language processing has produced datasets of Supreme Court judgments (ILSI, InLegalBert) and systems for case outcome prediction and legal question answering. Graph-based approaches have been applied to precedent citation networks.

The data quality challenges we document in Section 3.3 are not unique to this study. Damle and Anand (2020), in a systematic audit of eCourts metadata, found high rates of missing filing dates, malformed statute references, and unlinked disposal orders across multiple courts. Jain and Reddy (2025) demonstrate that the Bombay High Court's active caseload is overestimated by 66% because interlocutory applications are counted as separate primary cases. These prior audits establish that data quality failures in Indian judicial databases are structural rather than incidental, a theme that runs through our findings.

This paper fills the gap between aggregate trend analysis and individual case text analysis. We use the stage-level data available in the DAKSH dataset, combined with duration analysis, to build a structural picture of how different courts fail.

3. Data

3.1 Overview

This study uses two main data sources. The primary source is the DAKSH High Court Data Portal, which provides case-level records for writ petitions across Indian High Courts. The secondary source is the Indian Matrimonial Litigation Judgment Dataset (IMLJD), which covers criminal quash petitions from Karnataka High Court and the Supreme Court.

Both sources are publicly available and openly licensed. No proprietary data was used. The analysis is fully reproducible from public sources.

3.2 DAKSH High Court Data Portal

DAKSH India is a non-profit organisation that has collected case-level and hearing-level data from Indian High Courts through the eCourts mobile application and NJDG portal. The dataset is available at database.dakshindia.org under a CC BY-NC 4.0 licence after registration.

Each record in the DAKSH case-level file contains the following fields relevant to this study:

- CNR number (unique case identifier)
- Case type and category
- Date of filing and date of decision (where disposed)
- Current stage label
- Current status (pending or disposed)
- Hearing count
- Respondent name
- Court number and court name
- Pending days
- Disposal pattern (where recorded)

We downloaded the writ case CSV files for all available courts. Table 1 lists the ten courts used in this analysis. The dataset covers cases registered up to approximately 2021, reflecting the DAKSH collection timeline. Because structural institutional bottlenecks build up over years and decades, a dataset ending in 2021 remains representative of the structural failure types of these courts, though specific severity metrics may have changed since.

Table 1: DAKSH writ petition datasets used in this study

Court	File Size	Cases	Period
Allahabad HC	251 MB	820,114	Up to 2021
Andhra Pradesh HC	92 MB	255,322	Up to 2021
Calcutta HC	84 MB	272,322	Up to 2021
Chhattisgarh HC	69 MB	194,053	Up to 2021
Jammu and Kashmir HC	90 MB	251,556	Up to 2021

Karnataka HC	112 MB	367,275	Up to 2021
Kerala HC	35 MB	113,993	Up to 2021
Manipur HC	8 MB	20,498	Up to 2021
Meghalaya HC	3 MB	8,930	Up to 2021
Uttarakhand HC	51 MB	147,331	Up to 2021
Total		**2,451,398**	

3.3 Data Quality Problems

Before describing the findings, we document three systematic data quality problems in the DAKSH dataset. We describe them here rather than in footnotes because they affect any researcher using this data, and because honest characterisation of data quality is itself a contribution.

Problem 1: Sentinel values for missing filing dates. Karnataka and Andhra Pradesh both show a duration value of exactly 20,599 days (56.4 years) for a large proportion of cases. This is a sentinel value, meaning a placeholder stored in place of a missing or unknown filing date. It does not represent an actual case duration. Any analysis that does not detect and remove this sentinel will produce nonsensical duration statistics for these two courts. We detect it by checking for the specific value 20,599 and exclude affected records from duration analysis.

Problem 2: Non-UTF-8 encoding in the Jammu and Kashmir file. The J&K CSV file cannot be read with standard UTF-8 encoding because it contains regional language characters in party names. It requires latin-1 encoding. This is a minor technical issue but would cause a silent failure in any automated pipeline that does not handle it.

Problem 3: Stage label vocabulary is court-specific. Each court uses its own terminology for hearing stages. Calcutta uses "MOTION" for what other courts call "ADMISSION". Uttarakhand appends cause-list slot numbers to stage labels, producing entries like "ADMISSION MATTERS -25". Andhra Pradesh appends government department names to admission stages. These are not equivalent labels and cannot be compared directly. We describe our harmonisation approach in Section 4.2.

We also note that the disposal pattern field is empty for Allahabad and Karnataka, making attrition calculation impossible for those courts. The J&K disposal pattern field contains the typo "DISMISED" alongside "DISMISSED", creating a spurious vocabulary split that must be merged before analysis.

3.4 IMLJD Supplementary Dataset

The Indian Matrimonial Litigation Judgment Dataset (IMLJD, available at huggingface.co/datasets/joyboseroy/imljd) covers 3,613 Indian court judgments. For this study, we use only the Karnataka High Court sub-corpus of 2,139 cases covering criminal quash petitions filed under Section 482 of the Code of Criminal Procedure, decided between 2018 and 2024.

This subset provides clean registration and decision dates for Karnataka that are not available through DAKSH due to the sentinel value problem. It allows a partial duration analysis for Karnataka. Section 482 criminal petitions are procedurally simpler than constitutional writ petitions, are typically heard by a single judge, and are generally disposed faster. The duration figures from this subset are not comparable to the all-writ medians in Table 3 and should not be used to rank Karnataka against other courts.

3.5 What This Data Cannot Tell Us

The DAKSH case-level data does not include hearing-level transaction records. This means we cannot directly observe how cases move between stages, only what stage they are currently at. The bottleneck classification is therefore based on a snapshot of current stage distribution, not a trajectory or flow measurement.

The reserved judgment lag, the time between the last hearing and the delivery of the written judgment, is not directly observable in this dataset. We use the proportion of pending cases at the RESERVED stage as a proxy for Output bottleneck pressure, rather than measuring reservation lag directly.

For courts where the stage distribution is dominated by a single stage, the snapshot is highly informative. For courts with mixed distributions, the classification is less precise and should be treated with more caution.

4. Methods

4.1 Bottleneck Classification

For each court we compute the distribution of pending cases across canonical procedural stages. A court is classified by the stage that contains the largest share of its pending caseload, subject to minimum thresholds.

The thresholds draw on a basic queueing intuition. In a multi-stage service system, persistent accumulation at a stage indicates that the rate of downstream progression is lower than the rate of arrival at that stage. A court with 72% of pending cases at ADMITTED and unchanged for years is exhibiting a structural scheduling failure, not a random fluctuation. This logic applies the principle underlying Little's Law: the number of items waiting in a stage is proportional to the arrival rate times the average wait time at that stage. The specific cutoffs (45%, 50%, 15%) were chosen to reflect clear dominance rather than marginal plurality. Section 5.9 reports a sensitivity analysis.

We define four bottleneck classes:

Input bottleneck. More than 45% of pending cases are at the ADMITTED stage, or more than 80% of cases have zero recorded hearings. Cases are entering the system but not progressing to scheduled hearings.

Capacity bottleneck. More than 50% of pending cases are at the ARGUMENTS stage. Cases have been admitted and scheduled but there is not enough bench time to hear them all.

Output bottleneck. More than 15% of pending cases are at the RESERVED stage. Cases have been fully argued and reserved for a written judgment, but the judgment has not been delivered.

Process bottleneck. The intermediate evidence-gathering stage (filing of counter-affidavits, witness production, document production) dominates. This is a theoretically possible but empirically unobserved category in our dataset. Writ petitions are fought primarily on affidavits rather than through extended trial procedures. Process bottlenecks may be more common in civil suits and criminal trials, which are outside the scope of this study.

Where a court's distribution does not clearly fall into one class, we report it as mixed and describe the distribution directly.

4.2 Stage Harmonisation

The raw DAKSH stage labels cannot be compared across courts without harmonisation. We use a two-pass approach.

Pass 1 applies a rule-based mapping dictionary that converts raw labels to canonical stages: FILED, ADMITTED, NOTICED, COUNTER, ARGUMENTS, RESERVED, DECIDED, WITHDRAWN, and OTHER. The dictionary includes generic patterns as well as court-specific patterns for Calcutta, Karnataka, Andhra Pradesh, Uttarakhand, and Jammu and Kashmir, where the standard vocabulary does not apply.

For Uttarakhand, we strip trailing numeric slot identifiers before matching, converting "ADMISSION MATTERS -25" to "ADMISSION MATTERS". For Calcutta, we map "MOTION" and its variants to ADMITTED. For Andhra Pradesh, we map all variants of "ADMISSION (DEPARTMENT)" to ADMITTED regardless of the department suffix.

Pass 2 classifies any remaining unmatched labels as OTHER. We report the OTHER rate for each court as a data quality metric. Courts with OTHER rates above 40% are flagged as having stage vocabularies that are only partially decoded by our dictionary, and their bottleneck classifications are treated with correspondingly lower confidence.

4.3 Semantic Disorder Index

The Semantic Disorder Index (SDI) is a composite score capturing how fragmented and non-standard a court's stage label vocabulary is:

$$\mathbf{SDI = 0.4 \times H + 0.4 \times R + 0.2 \times S}$$

Where H is the Shannon entropy of the raw stage label distribution, R is the OTHER rate after harmonisation, and S is the proportion of pending cases at the ADMITTED stage with more than 10 recorded hearings. The weights are heuristic: entropy and unmappability are given equal weight as the primary signals of semantic fragmentation, with the process-stall indicator given half weight as a secondary signal. The index was not optimised against any outcome variable. It should be interpreted as one operationalisation of administrative encoding complexity among several possible alternatives.

As the findings show, SDI does not correlate with disposal rate. It measures registry encoding practice rather than court performance.

4.4 Attrition Signal

The attrition ratio is computed as the number of cases disposed by withdrawal, non-prosecution, or settlement divided by the number of cases disposed on merits (allowed, dismissed, or decided after hearing). We compute this from the DISPOSAL_PATTERN field where it is populated. The effective sample for attrition analysis is seven courts, after excluding Allahabad and Karnataka (field empty) and treating Calcutta (field sparse) as unreliable.

4.5 Government Respondent Classification

We classify a case as having a government respondent if the respondent name field contains any of a list of 30 patterns including "union of india", "state of", "government of", "district collector", "commissioner", "ministry of", "department of", and similar terms. This is an approximation; fuzzy name matching introduces some attenuation bias in downstream analysis. We validate it on a manual sample of 50 cases per court and find precision above 90% in all courts checked.

5. Findings

5.1 Bottleneck Classification

Table 2 summarises the bottleneck classification for all ten courts.

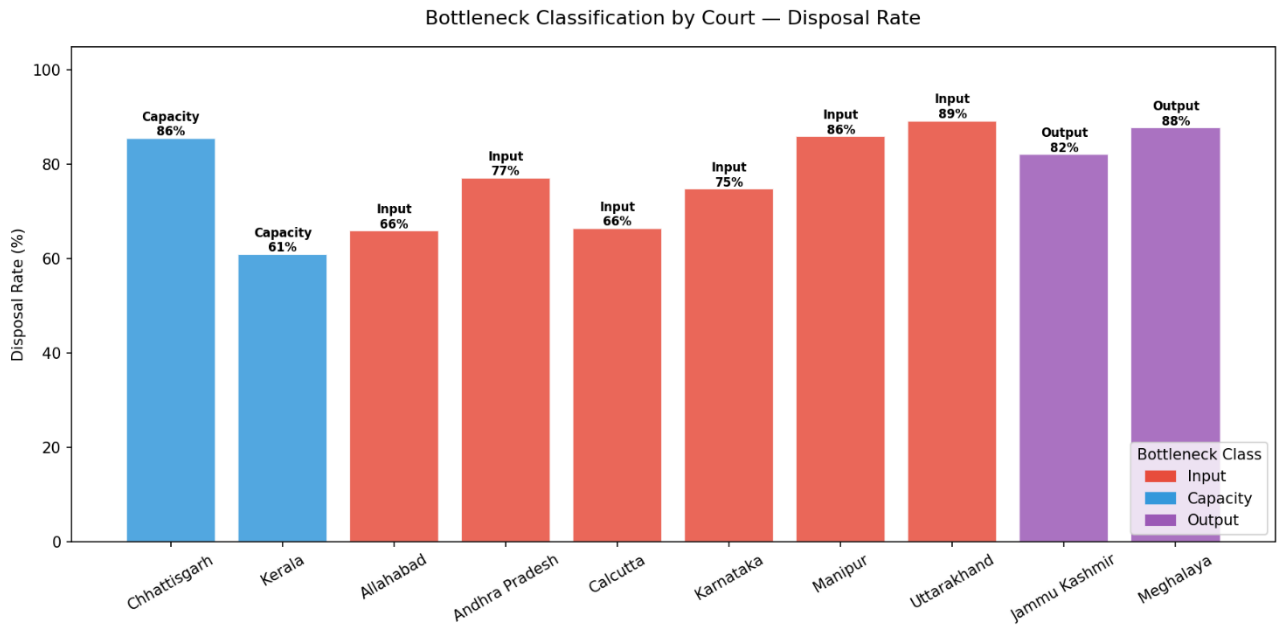

Figure 2: High Court Disposal Rates categorised by Structural Bottleneck Class. Bars indicate disposal rate labelled with the corresponding bottleneck type and dominant stage percentage (writ cases, DAKSH portal).

Table 2: Bottleneck classification across ten Indian High Courts

Court	Cases	Disposed%	Bottleneck	Dominant Stage (Pending)
Allahabad HC	820,114	66.0%	Input	ADMITTED 72.2%
Andhra Pradesh HC	255,322	77.2%	Input	ADMITTED 53.8%
Calcutta HC	272,322	66.5%	Input	ADMITTED 63.5%
Chhattisgarh HC	194,053	85.6%	Capacity	ARGUMENTS 61.4%
Jammu and Kashmir HC	251,556	82.2%	Output	ARGUMENTS 65.3%, RESERVED 25.7%
Karnataka HC	367,275	74.8%	Input	ADMITTED 46.5%
Kerala HC	113,993	61.0%	Capacity	ARGUMENTS 58.8%
Manipur HC	20,498	86.0%	Input	ADMITTED 59.2%
Meghalaya HC	8,930	87.8%	Output	RESERVED 47.8%
Uttarakhand HC	147,331	89.3%	Input	ADMITTED 50.3%

Six courts are Input bottleneck, two are Capacity, and two are Output. No court shows a Process bottleneck as the dominant failure mode.

A chi-square test of whether this distribution differs from a uniform spread across four classes gives a test statistic of 7.6 with a p-value of 0.055. With only ten courts and one of four theoretical classes unobserved, this sample does not provide strong statistical evidence for non-uniform clustering. We treat the bottleneck classification as descriptive and exploratory. The value of the classification lies in what it reveals about individual courts, not in the aggregate statistical test.

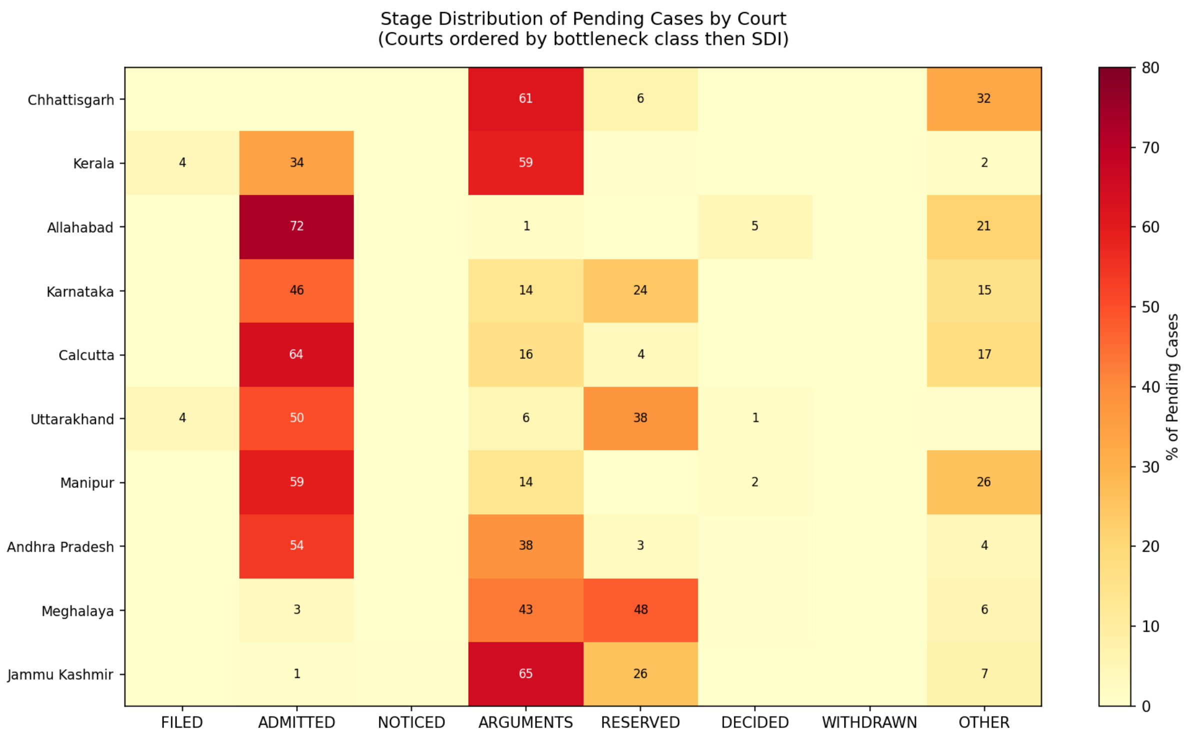

Figure 3: Heatmap of Pending Caseload Distribution Across Canonical Lifecycle Stages. Row values show the percentage of a court's pending backlog at each stage. Courts are ordered by bottleneck class and then by SDI.

5.2 Allahabad: The Largest Input Bottleneck

Allahabad High Court is the largest court in the sample with 820,114 writ cases. 72.2% of its pending cases are at the ADMITTED stage. The median case duration is 2,930 days, just over 8 years. 43.6% of cases have been pending for more than 10 years.

What is striking about Allahabad is that its stage vocabulary is the second smallest in the sample at 24 unique labels, and its SDI at 0.884 is the third lowest. This is a court that keeps orderly administrative records of cases that are not moving. The admission queue is clear and well-documented. The problem is structural: there is no pathway from the admission stage to a scheduled hearing at a rate that keeps up with new filings.

Government respondents appear in 58.3% of Allahabad cases, the second highest rate in the sample. A large share of the pending caseload involves the Uttar Pradesh state government or central government agencies as respondents.

5.3 Calcutta: Vocabulary Fragmentation and Institutional Paralysis

Calcutta High Court has 272,322 cases, a disposal rate of 66.5%, and a median case duration of 3,792 days (10.4 years). It is the slowest court in the sample on duration.

Calcutta uses 350 unique stage labels, more than 21 times as many as Meghalaya. The vocabulary includes extensive sub-categories of the basic admission hearing: "MOTION", "NEW MOTION", "LISTED MOTION", "MOTION 1", "MOTION 2", "CIVIL NEW MOTION", "URGENT MOTION". These all describe variations of what other courts call an admission hearing. Calcutta's registry has developed its own internal dialect, entirely different from the terminology used anywhere else.

The most alarming statistic for Calcutta is the zero-hearing rate: 93.7% of cases have no recorded hearings. This may reflect genuine scheduling failure, but it may also partly reflect registry recording practices that differ from other courts. A check of disposed Calcutta cases would help distinguish these explanations, but is not possible with the current DAKSH data because the disposal pattern field is also sparsely populated for Calcutta. Both the zero-hearing rate and the low disposal pattern coverage point to deep data infrastructure problems that make Calcutta the hardest court to analyse in this sample.

Government respondent share at 69.1% is the highest in the sample.

5.4 Meghalaya and Jammu and Kashmir: The Output Bottleneck

Meghalaya and J&K are the two Output bottleneck courts. Both have high disposal rates (87.8% and 82.2%), low zero-hearing rates (32.7% and 2.0%), and large proportions of pending cases at the RESERVED stage (47.8% and 25.7%). Cases in these courts are being heard. The constraint is in the writing stage.

This is a different institutional failure from the Input and Capacity courts. More judges would not help. The constraint is in judgment delivery, which happens outside court hours and does not depend on the number of judges available to hear new cases. It would not show up in any analysis that only looks at aggregate pendency counts.

Meghalaya has the highest attrition ratio in the sample at 0.527, meaning that for every two cases disposed on merits, one is withdrawn. A court with a large RESERVED backlog may be generating pressure on litigants to settle rather than wait for judgment.

An unusual feature of the J&K data is that 5,264 pending cases, roughly 12% of all pending cases, are classified under the stage "CASES TO BE TRANSFERRED TO CENTRAL ADMINISTRATIVE TRIBUNAL". These cases are pending not because of any hearing queue or judgment delivery failure, but because an administrative transfer order has not been executed. They have no judicial content. Including them in any pendency count inflates the J&K figure without reflecting judicial capacity or performance. Administrative processing states should be structurally filtered out of judicial performance metrics.

5.5 Chhattisgarh and Kerala: Capacity Bottleneck

Chhattisgarh and Kerala are the two Capacity bottleneck courts. Their pending caseloads are dominated by the ARGUMENTS stage (61.4% and 58.8% respectively), meaning cases have been admitted and scheduled but are waiting for bench time.

Chhattisgarh's median duration of 146 days is the shortest in the sample. Only 7.5% of cases have been pending for more than 10 years. The Capacity bottleneck in Chhattisgarh appears manageable: the court is keeping pace despite a large arguments queue.

Kerala's profile is more concerning. Median duration of 700 days (1.9 years) and 24.1% pending over 10 years. Kerala's stage vocabulary is the cleanest in the sample among large courts: 54 labels, 1.2% OTHER rate, SDI of 1.113. Kerala's registry records are reliable. The problem is bench capacity, not administrative dysfunction.

5.6 Duration Heterogeneity

Table 3 shows median case duration for the eight courts where reliable duration data is available. Karnataka and Andhra Pradesh are excluded due to the sentinel value problem in the DAKSH data.

Table 3: Median case duration by court (DAKSH data, valid courts only)

Court	Median Days	Median Years	Bottleneck	Note
Chhattisgarh HC	146	0.4	Capacity	
Meghalaya HC	162	0.4	Output	
Uttarakhand HC	195	0.5	Input	
Manipur HC	328	0.9	Input	
Kerala HC	700	1.9	Capacity	
J&K HC	1,013	2.8	Output	
Allahabad HC	2,930	8.0	Input	
Calcutta HC	3,792	10.4	Input	Slowest

The 26-fold range from 146 to 3,792 days is the paper's most direct evidence for court-level heterogeneity. Chhattisgarh and Calcutta operate under identical constitutional and procedural frameworks. One resolves the median writ petition in under five months. The other takes more than ten years. No single national policy can simultaneously address both.

5.7 Semantic Disorder Index

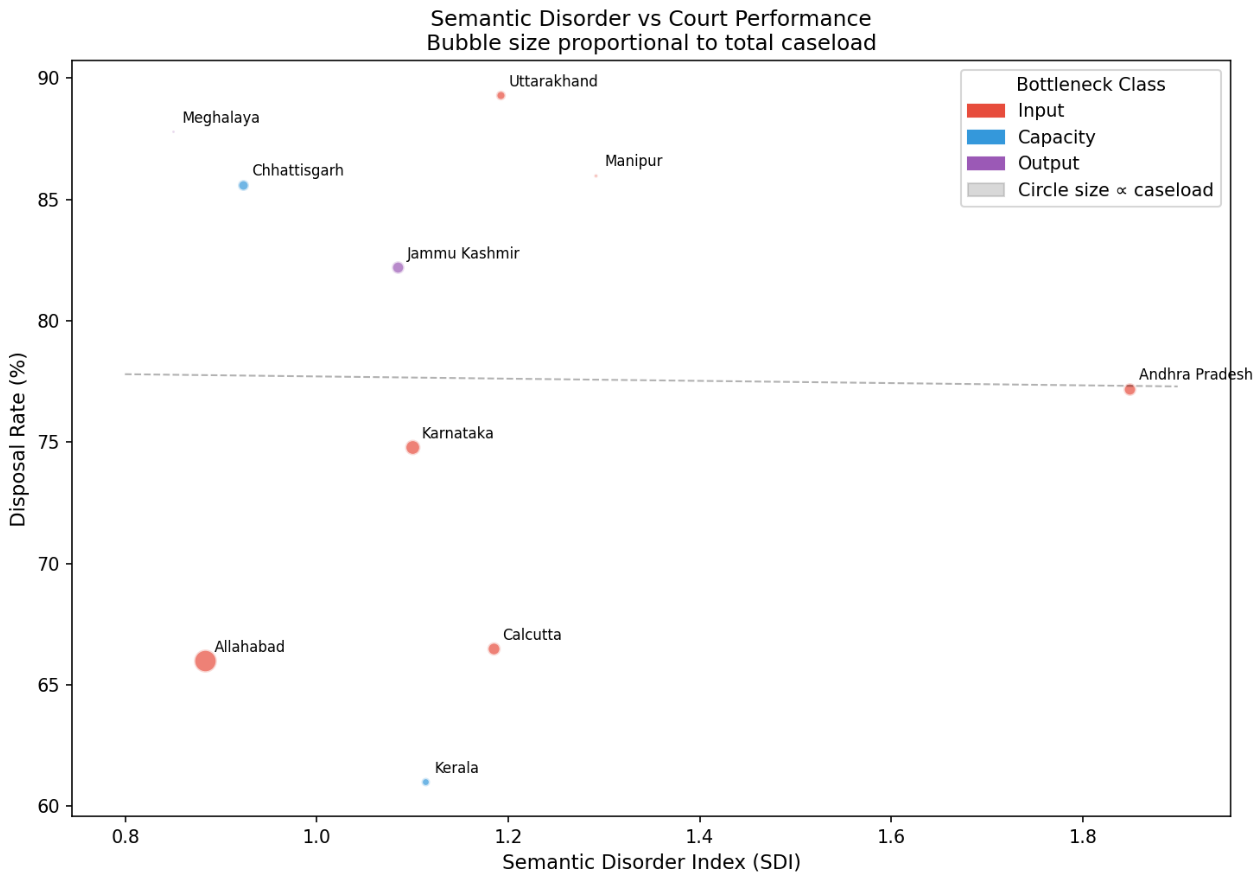

Figure 4: SDI vs. Court Disposal Rate. Bubble size is proportional to total writ caseload. The dashed line indicates a lack of linear correlation. SDI captures registry encoding styles rather than operational efficiency.

Table 4 shows the SDI for all ten courts.

Table 4: Semantic Disorder Index by court

Court	SDI	Vocab Size	OTHER Rate	Bottleneck	Note
Meghalaya HC	0.850	16	62.5%	Output	Smallest vocab
Allahabad HC	0.884	24	40.3%	Input	
Chhattisgarh HC	0.923	33	69.1%	Capacity	
J&K HC	1.085	35	3.0%	Output	Lowest OTHER rate
Karnataka HC	1.100	125	21.0%	Input	
Kerala HC	1.113	54	1.2%	Capacity	Cleanest large court
Calcutta HC	1.185	350	69.7%	Input	Largest vocab

Uttarakhand HC	1.191	100	8.2%	Input	Slot numbers inflate SDI
Manipur HC	1.291	37	50.3%	Input	
Andhra Pradesh HC	1.849	243	5.5%	Input	Dept names inflate SDI

The SDI does not predict disposal rate (Spearman r = 0.055, p = 0.881). It measures registry encoding practice rather than performance. Courts with high SDI have embedded additional information in stage labels: scheduling slots in Uttarakhand, government department names in Andhra Pradesh. This inflates vocabulary and entropy without reflecting procedural complexity.

J&K (SDI 1.085, 35 labels) illustrates what a well-structured vocabulary looks like: descriptive stage names like "FOR ADMISSION (After Notice)" that encode procedural state precisely without fragmentation. Andhra Pradesh (SDI 1.849, 243 labels) illustrates the opposite: a vocabulary that has grown by accretion of department-specific variants of a single stage, making cross-court comparison impossible without harmonisation.

5.8 Attrition Signal

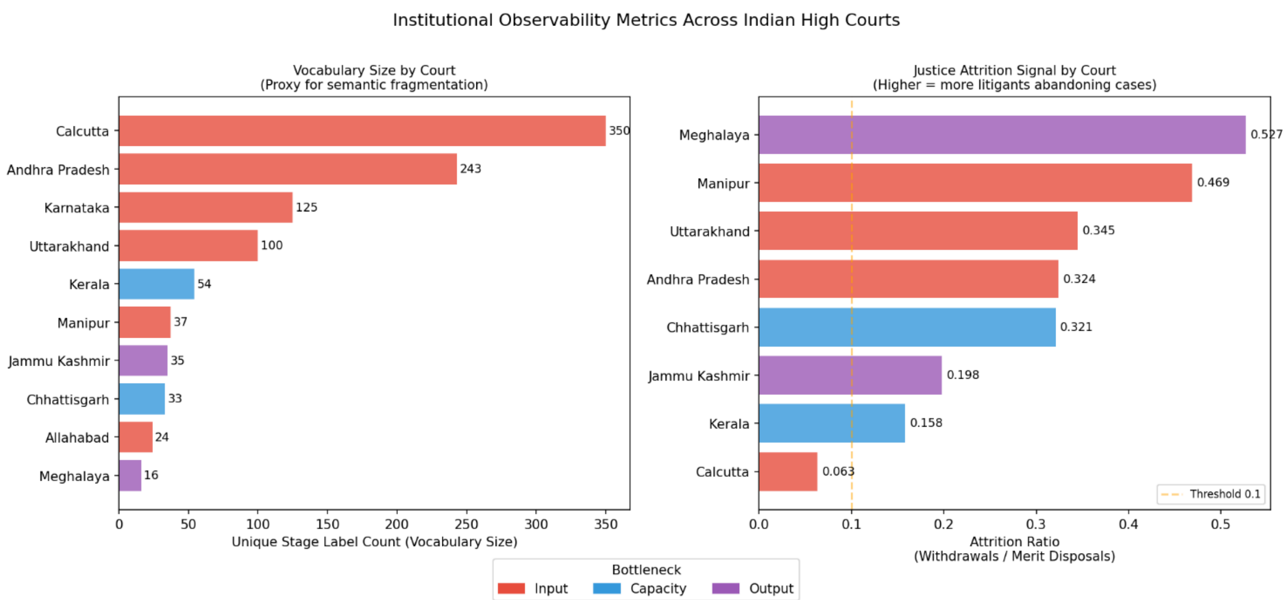

Figure 5: Vocabulary Size vs. Attrition Ratio. Left panel: unique stage label count as a proxy for semantic fragmentation. Right panel: attrition ratio (withdrawals divided by merit disposals). Courts where delay correlates with case exit via litigant exhaustion are highlighted.

Table 5 shows the attrition ratio for courts where disposal pattern data is available. The effective sample is seven courts after excluding Allahabad and Karnataka.

Table 5: Attrition ratio by court

Court	Withdrawal Rate	Merit Disposal Rate	Attrition Ratio	Note
Meghalaya HC	7.5%	3.8%	0.527	Highest
Manipur HC	6.7%	3.8%	0.469	
Uttarakhand HC	3.5%	8.9%	0.345	
Andhra Pradesh HC	3.0%	6.9%	0.324	
Chhattisgarh HC	4.2%	8.9%	0.321	
J&K HC	2.7%	6.9%	0.198	
Kerala HC	1.8%	4.5%	0.158	
Calcutta HC	0.5%	1.1%	0.063	Data sparse
Allahabad HC	n/a	n/a	n/a	Field empty in DAKSH
Karnataka HC	n/a	n/a	n/a	Field empty in DAKSH

Meghalaya has the highest attrition ratio at 0.527, Manipur at 0.469. Both courts have government respondent shares above 70%. Among the seven courts with reliable data, government respondent share and attrition ratio show a directional positive correlation (Spearman r = 0.535, n = 7). Given the small sample, this is an exploratory signal rather than a statistically robust result. It is consistent with the hypothesis that litigants facing government respondents are more likely to abandon cases before adjudication, but alternative explanations including local litigation culture and settlement norms cannot be ruled out.

Calcutta's very low rates reflect data absence. The vast majority of disposed Calcutta cases have no recorded disposal nature, not genuine low withdrawal.

5.9 Sensitivity Analysis of Classification Thresholds

Table 7 shows how the bottleneck classification changes when each threshold is varied by plus or minus 10 percentage points. Nine of the ten courts are stable across all threshold variations. Karnataka is the only court whose classification changes: at a threshold of 35% ADMITTED (the base minus 10 percentage points), it would be classified as mixed rather than Input, since its pending ADMITTED share is 46.5%, just above the base threshold of 45%.

Table 7: Sensitivity of bottleneck classification to threshold variation

Court	Base Thresholds	Thresholds +10pp	Thresholds -10pp	Stability
Allahabad HC	Input	Input	Input	Stable
Andhra Pradesh HC	Input	Input	Input	Stable
Calcutta HC	Input	Input	Input	Stable

Chhattisgarh HC	Capacity	Capacity	Capacity	Stable
Jammu and Kashmir HC	Output	Output	Output	Stable
Karnataka HC	Input	Input	Mixed	Changes at -10pp
Kerala HC	Capacity	Capacity	Capacity	Stable
Manipur HC	Input	Input	Input	Stable
Meghalaya HC	Output	Output	Output	Stable
Uttarakhand HC	Input	Input	Input	Stable

The finding of heterogeneous bottleneck types is robust to the threshold choices. The borderline case of Karnataka at 46.5% ADMITTED is noted explicitly in the text wherever Karnataka's classification is discussed.

6. Supplementary Evidence

6.1 Karnataka: Evidence from IMLJD

The DAKSH data cannot be used for duration analysis in Karnataka. To provide a partial estimate, we use 2,137 criminal quash petitions (Section 482 CrPC) from the Karnataka HC IMLJD sub-corpus, decided between 2018 and 2024.

The median duration is 229 days. This is not comparable to the all-writ medians in Table 3 because Section 482 petitions are procedurally simpler, heard by a single judge, and typically resolved faster than constitutional writ petitions. Table 6 shows the duration by filing year.

Table 6: Karnataka HC Section 482 petitions, median duration by filing year

Filing Year	Cases	Median Duration (Days)	Note
2018	(included)	282	
2019	(included)	1,211	COVID effect: 4.5x spike
2020	(included)	728	Partial recovery
2022	(included)	268	Post-COVID recovery
2023	(included)	127	Near pre-COVID levels
2024	(included)	252	

The 2019 cohort has a median of 1,211 days, approximately four times longer than any other year. This is the COVID effect: cases filed in 2019 were mid-stream when physical court operations suspended in March 2020. The 2022 and 2023 cohorts show recovery to sub-300-day medians.

This data contributes one specific finding: case duration is sensitive to exogenous shocks in ways that static extrapolative models cannot account for. A linear regression fitted to 2017-2019 data would not predict the

2019-2021 spike or the 2022 recovery. This is an additional argument against deterministic pendency clearance projections.

6.2 The CAT Transfer Cases in J&K

5,264 pending J&K cases (about 12% of all pending cases) are classified under "CASES TO BE TRANSFERRED TO CENTRAL ADMINISTRATIVE TRIBUNAL". These are pending because an administrative transfer order has not been executed, not because of any judicial queue.

This illustrates a broader problem with pendency counts. They aggregate cases in fundamentally different situations: cases waiting for a first hearing, cases waiting for a judgment, and cases waiting for administrative processing. National pendency statistics that combine all three categories cannot support meaningful reform design.

7. Discussion and Policy Implications

7.1 The Core Finding

Indian High Courts fail in structurally different ways. Six courts have pending caseloads concentrated at the admission stage. Two have them concentrated at the arguments stage. Two have them concentrated at the reserved stage. These are three different problems requiring three different interventions.

Input bottleneck courts need front-end interventions: better docket management, faster admission scheduling, or demand-side reforms reducing filings. Capacity bottleneck courts need middle interventions: more bench time, better calendar management, or faster hearings. Output bottleneck courts need back-end interventions: mandatory timelines for reserved judgment delivery, electronic tracking of pending judgments, administrative alerts when reservation periods exceed defined limits.

7.2 What Vacancy Filling Will and Will Not Do

Vacancy filling is the dominant reform prescription. It is correctly targeted for Capacity and Input bottleneck courts, where the binding constraint is bench time or scheduling capacity.

But vacancy filling will have no effect on Output bottleneck courts. Meghalaya and J&K have a 2.0% and 32.7% zero-hearing rate respectively. Cases are being heard. The problem is in judgment delivery, which happens outside court hours. Appointing additional judges to Meghalaya will not accelerate the delivery of judgments that have already been reserved.

This is a direct and testable implication. If judicial vacancy filling improves disposal rates in Output bottleneck courts, the classification is wrong. If it does not, the classification is right.

7.3 The Data Infrastructure Problem

A secondary finding is that the data infrastructure for Indian judicial research is inadequate for structural analysis. The DAKSH dataset contains sentinel values making duration analysis impossible for two courts, court-specific encoding requiring manual harmonisation, and a disposal pattern field empty for the two largest courts. The NJDG does not include a standardised stage label taxonomy. The delay reason field that exists in the district court NJDG does not exist at the High Court level.

The administrative processing states in the J&K data, cases pending only because a transfer order has not been executed, point to a further problem: pendency counts do not distinguish judicial delay from administrative delay. This conflation makes both courts appear worse than they are on judicial efficiency metrics.

7.4 Minimum Dataset Specification

We identify five minimum data requirements for the structural analysis this paper performs and for the more granular analysis that would improve it:

- A standardised national stage label taxonomy, maintained by the eCommittee of the Supreme Court, applied uniformly across all High Courts and updated annually.
- A delay reason field at the High Court level, with a restricted controlled vocabulary mapping to four causal classes: court-attributable, counsel-attributable, party-attributable, and investigation-attributable.
- The reservation date as a separate field from the last hearing date, so that reserved judgment lag can be computed directly.
- Reliable filing dates for all cases, with explicit NULL values where unknown rather than sentinel placeholders.
- A disposal nature field populated for all disposed cases, not only a subset.

A sixth requirement emerges from the J&K finding: an administrative processing status flag that distinguishes cases pending for judicial reasons from cases pending for administrative reasons such as unexecuted transfers. Including administrative cases in pendency counts inflates figures and distorts performance assessment.

All six requirements are technically feasible within the existing CIS infrastructure. Requirements 1, 3, 4, and 5 require only validation changes in data entry. Requirement 2 mirrors field infrastructure that already exists at the district court level. None requires new data collection from parties or advocates.

8. Conclusion

We analyse 2.45 million writ petition records from ten Indian High Courts and find that courts fail in structurally different ways. Six courts show Input bottlenecks, two show Capacity bottlenecks, and two show Output bottlenecks. Median case duration varies 26-fold. Stage vocabulary varies 22-fold. Neither variation is random.

The implication is straightforward. Judicial reform designed at the national level cannot address problems that are heterogeneous at the court level. A reform that addresses the right constraint in Allahabad may be irrelevant in Meghalaya, and vice versa. Effective reform requires diagnostic precision before prescriptive action.

The findings in this paper are based on writ petitions only. Civil suits, criminal trials, and tribunal matters may exhibit different bottleneck patterns and are a priority for future work. The data covers cases up to 2021, and structural bottleneck types may have shifted since. The bottleneck classification should be treated as descriptive and exploratory rather than definitively established.

The data infrastructure to support this kind of structural diagnosis does not currently exist in the form needed. We propose six specific additions to the NJDG and eCourts data schema that would make structural diagnosis tractable across all tiers of the Indian judiciary. We encourage researchers, civil society organisations, and judicial administration bodies to extend this analysis to additional courts, additional case types, and future years as the DAKSH dataset is updated.

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Appendix A: Stage Harmonisation Dictionary

The complete mapping from raw DAKSH stage labels to canonical stages is in the analysis code. The court-specific overrides for Calcutta, Karnataka, Andhra Pradesh, Uttarakhand, and Jammu and Kashmir are documented in the COURT_SPECIFIC_PATTERNS dictionary in that file. The code is available to researchers on request from the author.

Appendix B: Data Quality Notes by Court

Full data quality notes including null rates, sentinel value counts, encoding issues, and OTHER rates for each of the ten courts are in outputs/cross_court_summary.csv in the analysis code package. Available to researchers on request.

Appendix C: Replication

All analysis was conducted using Python 3.10 with pandas, numpy, scipy, and matplotlib. The DAKSH dataset is available at database.dakshindia.org under CC BY-NC 4.0. The IMLJD dataset is available at huggingface.co/datasets/joyboseroy/imljd under CC BY 4.0. Analysis code is available from the author on request.